

The Basics

O₂ Devices

Cases

*Click on any rounded button with a
cursor to navigate to other pages*

The Basics

O₂ Devices

Cases

Respiratory Physiology

Oxygenation

↑ FiO₂

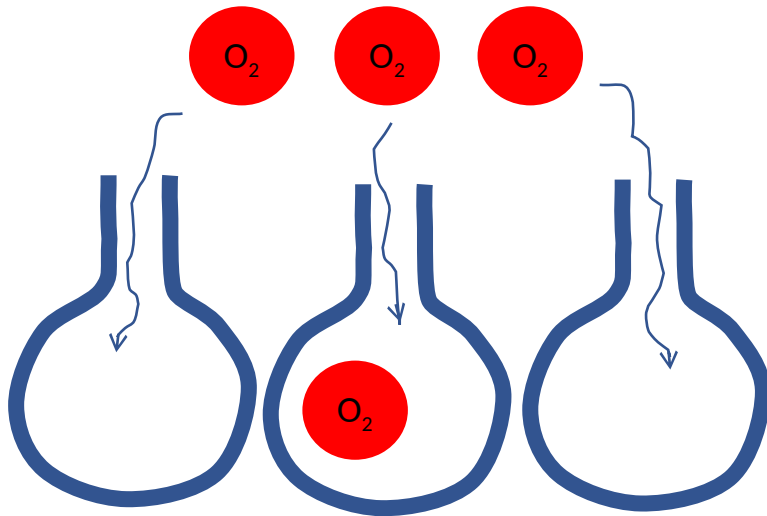
↑ number or % of O₂ molecules
participating in gas exchange
(room air is 21% FiO₂)

↑ PEEP

Recruits additional alveoli
during expiration to
participate in gas exchange

Ventilation

Tidal volume (V_t) x Respiratory rate



The Basics

O₂ Devices

Cases

Oxygen Delivery Systems

Oxygenation

Ventilation

Nasal cannula

Face mask (simple)

Oxymizer

Non-rebreather mask

Venturi mask

High flow nasal cannula

CPAP

BiPAP

Mechanical ventilation

↑ max FiO₂
↓ 100%

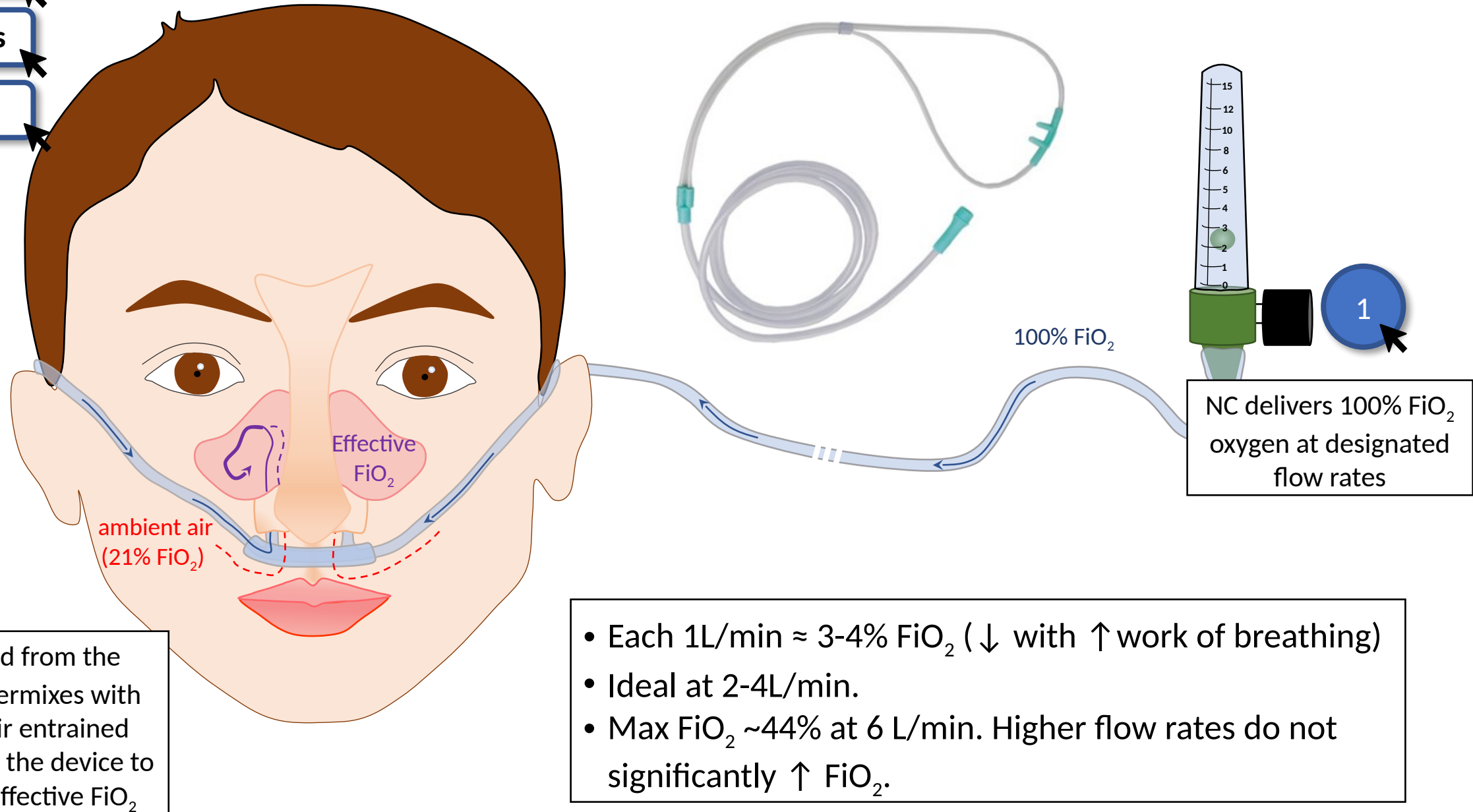
PEEP

The Basics

O₂ Devices

Cases

Nasal Cannula

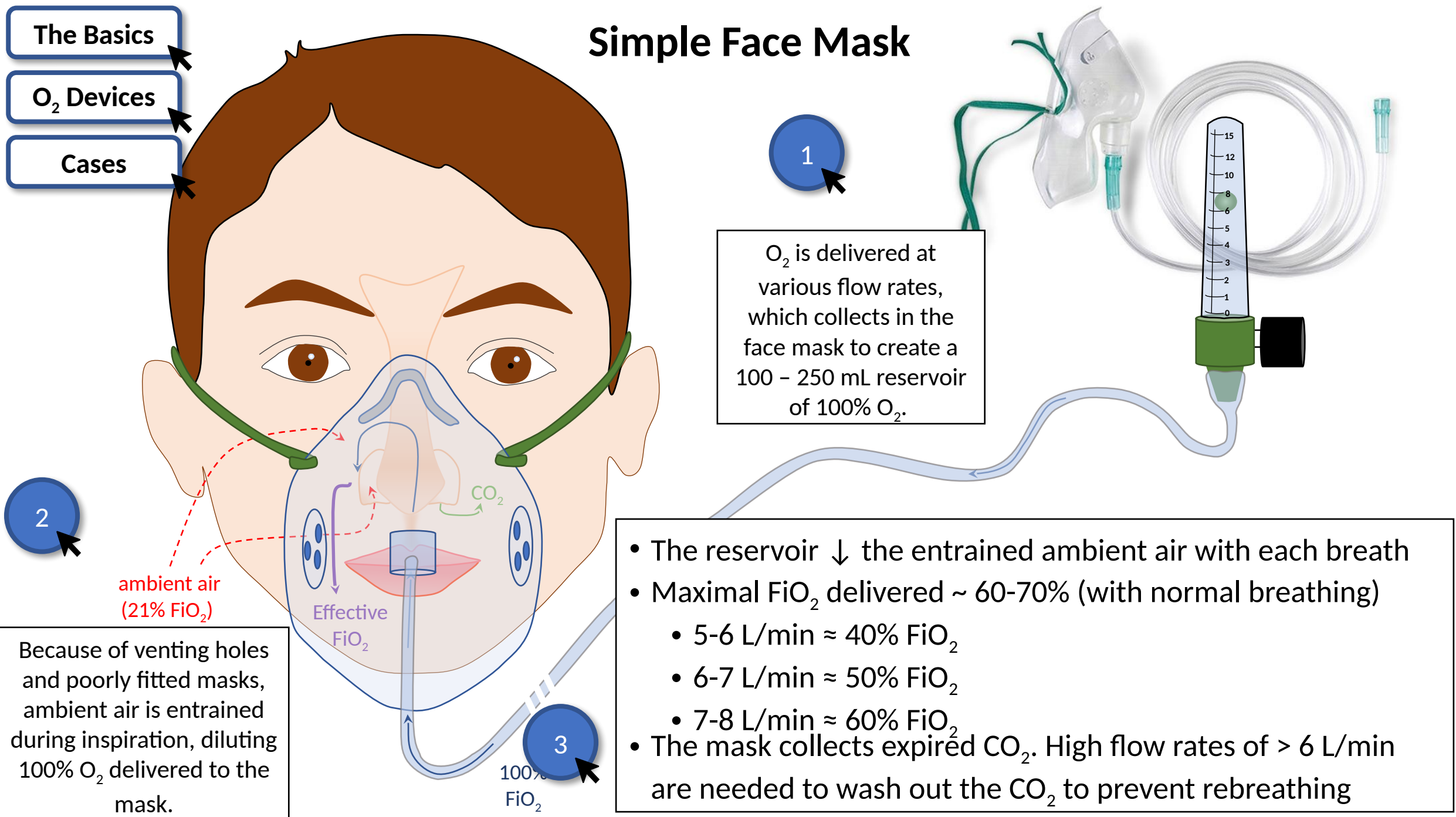


The Basics

O₂ Devices

Cases

Simple Face Mask

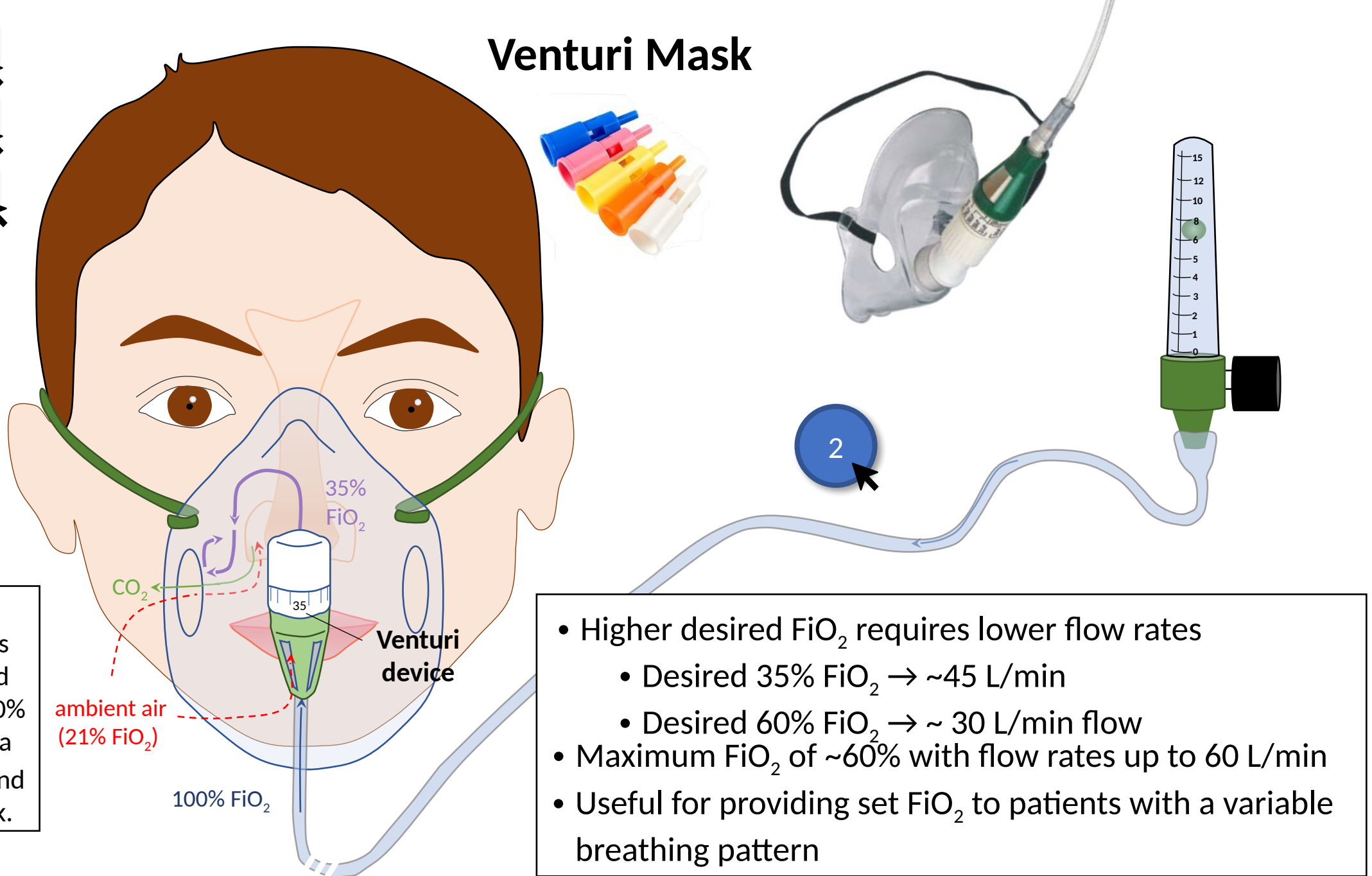


The Basics

O₂ Devices

Cases

Venturi Mask



The Basics

O₂ Devices

Cases

High Flow Nasal Cannula

2

Thick nasal
cannula prongs

High flow rates of O₂ wash out the nasopharynx and upper airways with the delivered O₂, creating an oxygen reservoir. This also ↓ CO₂ rebreathing and dead space.

PEEP

Trachea
Set FiO₂

3

Thicker nasal cannula prongs, which occlude the nasal passages, combined with the high flow rates, generate small amounts of PEEP (~ 3-5 cm H₂O).

Humidifier

- Adjustable FiO₂ and flow rate.
- Maximum flow rate is ~ 60 L/min and maximum effective FiO₂ is near 100%.
- Higher flow rates delivered are able to match higher inspiratory flow rates in patients with ↑ work of breathing and minute ventilation.

1

The Basics

O₂ Devices

Cases

Non-rebreather Mask



ambient air
(21% FiO₂)

One way valve
(exhalation)

One way valve
(inhalation)

A one-way inhalation valve between the mask and reservoir bag allows O₂ to flow from the tubing → reservoir bag → mask but does not allow exhaled CO₂ to re-enter the bag. Two-way exhalation valves let out exhaled CO₂.

100% FiO₂

Reservoir bag

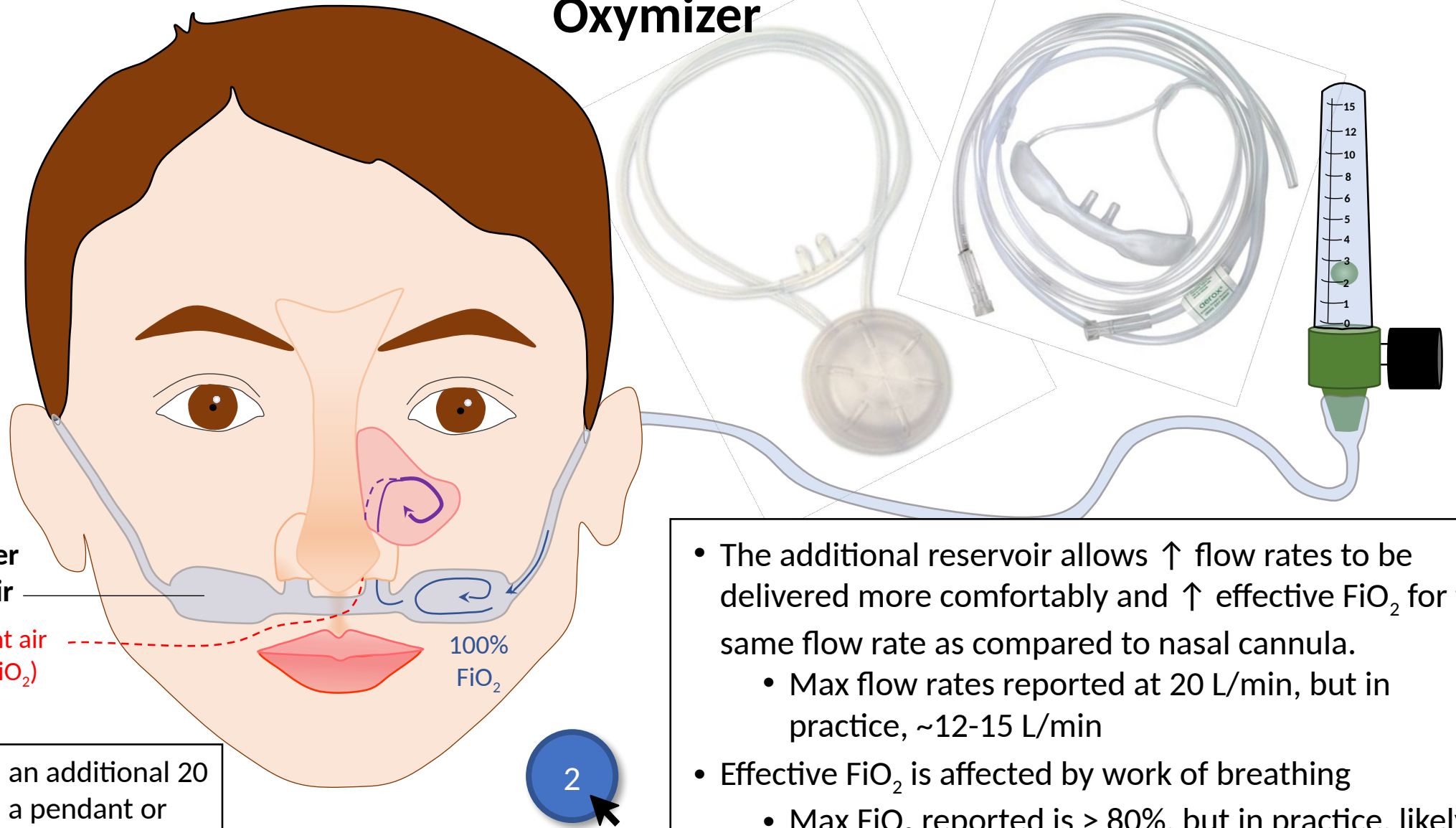
- One-way valves control the flow of O₂ and CO₂ to allow the patient to preferentially breathe 100% FiO₂ from the reservoir bag and expel CO₂ into the environment.
- Maximum effective FiO₂ of 75 – 90% at flow rates of 12 – 15 L/min.
- **Rescue form of oxygen only!!**

The Basics

O₂ Devices

Cases

Oxymizer



**Oxymizer
reservoir**

ambient air
(21% FiO₂)

100%
FiO₂

Oxymizers contain an additional 20 mL reservoir in a pendant or "mustache" shape.

- The additional reservoir allows ↑ flow rates to be delivered more comfortably and ↑ effective FiO₂ for the same flow rate as compared to nasal cannula.
 - Max flow rates reported at 20 L/min, but in practice, ~12-15 L/min
- Effective FiO₂ is affected by work of breathing
 - Max FiO₂ reported is > 80%, but in practice, likely lower

*maximum values noted are reported from manufacturing details

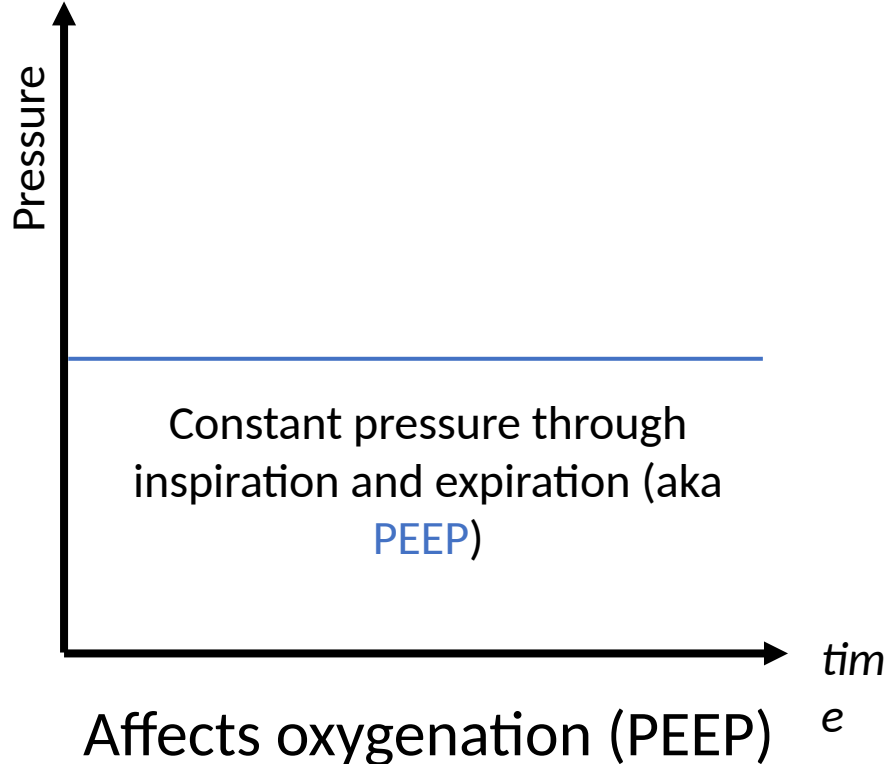
Dumont, CP & Tiep, BL. 2002. Using a Reservoir Nasal Cannula in Acute Care. Critical Care Nurse. 22(4): 41-46.

Tiep, BL & Belman, MJ. 1985, A new Pendant Storage Oxygen-conserving Nasal Cannula. CHEST. 87: 381-383.

Noninvasive Positive Pressure Ventilation

Allows for titration of both FiO_2 and PEEP

CPAP



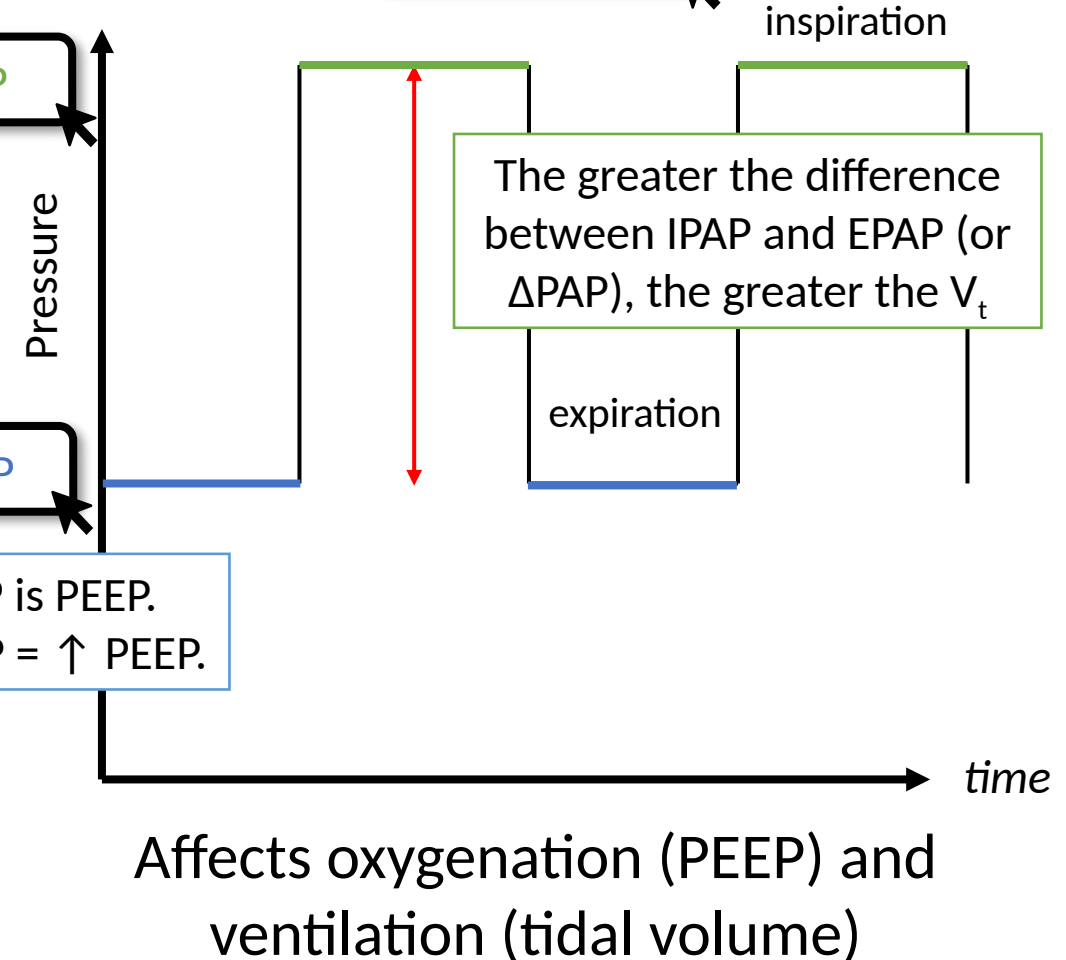
IPAP recruits additional alveoli during inspiration $\rightarrow \uparrow V_t$

IPAP

EPAP

EPAP is PEEP.
 $\uparrow \text{EPAP} = \uparrow \text{PEEP}$.

BiPAP



Noninvasive Positive Pressure Ventilation

Physiologic Effects

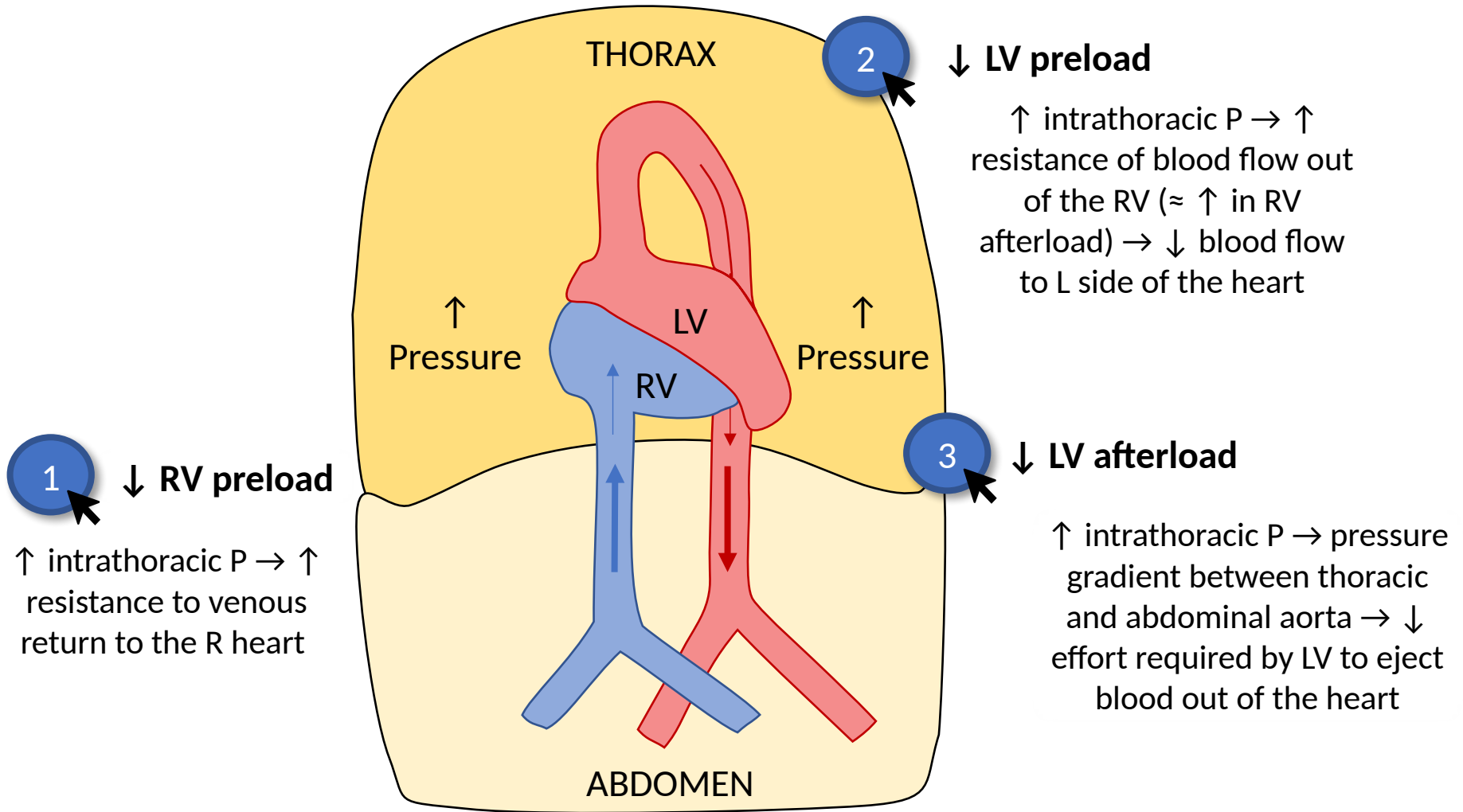
The Basics

O₂ Devices

NIPPV Effects

Indications

Cases



Noninvasive Positive Pressure Ventilation

Indications in acute respiratory failure

CPAP

BiPAP

The Basics

O₂ Devices

NIPPV Effects

Indications

Cases

ACUTE
RESPIRATORY
FAILURE

Acute cardiogenic pulmonary edema

COPD exacerbation with
hypercapnic respiratory failure

Immunocompromised patients
Post-operative respiratory failure
Post-extubation respiratory failure
Palliative

CHRONIC

OSA/ OHS

Neuromuscular disease

Contra-
indications

Inability to wear a mask (facial trauma)

Active nausea/vomiting, copious secretions

Inability to pull off a mask (AMS, paralysis)

Recent esophageal or tracheal surgery

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O₂ Devices

Cases

Case 1

64 yo with R sided renal cell carcinoma present for a R nephrectomy. On POD5, they develop acute onset SOB. VS: T 37.1C, HR 140, BP 90/60, RR 28, SaO₂ 82% on RA.

Trial of nasal cannula can be used for mild-moderate hypoxemia

They progressively desaturate and are now on 6L NC with SaO₂ 89%.

They can be escalated to an oxymizer, Venturi mask or HFNC

On POD7, they are on HFNC 100% FiO₂, 60 L/min. RR is 24 and shallow, SaO₂ 86%. They are obtunded and difficult to arouse. ABG shows 7.21/72/45/20.

Intubation and mechanical ventilation.

There is hypoxemic and hypercarbic respiratory failure despite max FiO₂. PEEP is needed to ↑ PaO₂. BiPAP is contraindicated with AMS. Other contraindications to NIPPV: facial trauma, copious secretions, nausea/vomiting, hemodynamic instability.

Case 2

74 yo with a history of COPD presents with a COPD exacerbation. VS: T 37.1C, HR 120, BP 160/60, RR 32, SaO₂ 82% on RA. PE: Sleepy but easily arouses. Diffuse wheezing. CXR is clear. ABG shows 7.21/72/45/28.

BiPAP is needed for hypoxemic and hypercarbic respiratory failure.

They are placed on BiPAP of 10/5, 35% FiO₂. Repeat ABG shows 7.30/60/79/28.

Increase the IPAP. There is persistent hypercarbic respiratory failure. To improve ventilation, ↑ the ΔPAP.

There is persistent hypoxemia (PaO₂ 55) but respiratory acidosis/ hypercapnia is resolved (7.35/45). Hypoxemia can be addressed by ↑ FiO₂ or EPAP/IPAP proportionately (e.g., 12/7)

There is persistent hypoxemia (PaO₂ 55) and hypercapnia (7.30/60). Options include:

- ↑ IPAP alone + ↑ FiO₂. E.g. 12/5, 50% FiO₂
- Or ↑ EPAP and ΔPAP. E.g., 15/7.

Case 3

56 yo with HFrEF is admitted with a HFrEF exacerbation. CXR shows diffuse bilateral alveolar infiltrates. VS: T 36.5C, HR 110, BP 160/80, RR 32, SaO₂ 75% on RA.

Given severity of hypoxemia, consider a **non-rebreather** to quickly improve oxygenation.

On non-rebreather, SaO₂ improves to 89%, RR is now 28. ABG shows 7.46/29/59/24.

Start CPAP. They have persistent hypoxemia despite near max FiO₂. Additionally, acute cardiogenic pulmonary edema is a strong indication for NIPPV.

Oxygen Delivery — 2022–25 Context Update

The device physiology, NIPPV mechanics, and indications are timeless and verified current (June 2026). The items below add modern evidence; standing teaching unchanged.

HFNC first-line for acute hypoxemic respiratory failure: ESICM ARDS 2023 recommends HFNC over conventional oxygen in non-intubated AHRF to reduce intubation — EXCEPT cardiogenic pulmonary edema and COPD exacerbation, where NIPPV (CPAP/BiPAP) is preferred. Consistent with FLORALI (Frat, NEJM 2015).

Monitor HFNC with the ROX index = $(\text{SpO}_2/\text{FiO}_2) \div \text{RR}$. $\text{ROX} \geq 4.88$ at 2/6/12 h predicts HFNC success; a low or falling ROX flags impending failure — escalate, do not delay intubation (Roca, AJRCCM 2019).

Awake prone positioning is a recommended adjunct for non-intubated patients with COVID-related hypoxemic AHRF on HFNC/NIPPV — reduces intubation (strong rec, 2023 rapid practice guideline); trial 15 min, continue if tolerated.

CPAP vs HFNC in COVID AHRF (RECOVERY-RS, JAMA 2022): CPAP reduced the intubation/death composite vs conventional oxygen; HFNC did not differ. HFNC and NIV are otherwise similarly effective at avoiding intubation in AHRF (2023 meta-analysis).

Take-Home Points

1. Hypoxemia is corrected by raising FiO₂ or adding PEEP. Most ward devices (nasal cannula, simple/Venturi/NRB masks, oxymizer) adjust only FiO₂; effective FiO₂ falls as work of breathing/inspiratory flow rises.
2. HFNC delivers high, matched flow with finely titratable FiO₂ (up to ~100%) and a small, non-adjustable PEEP (~3-5 cm H₂O); it improves oxygenation, work of breathing, and minute ventilation.
3. NRB is a RESCUE device only — de-escalate once stable. Venturi gives a precise set FiO₂ for variable breathing patterns.
4. CPAP = oxygenation only (PEEP). BiPAP (IPAP–EPAP = Δ PAP) adds ventilation → use for hypercapnic failure. Both unload the heart (↓ RV/LV preload, ↓ LV afterload) — strong in acute cardiogenic pulmonary edema; BiPAP for COPD hypercapnic failure.
5. First-line for AHRF is HFNC (except cardiogenic edema and COPD); add awake proning; track the ROX index and escalate early.

Attribution & Optimization Notes

Original lesson: "Oxygen Delivery Systems," TeachIM.org (updated Oct 2021).

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Corrections made (this edit, all single-run XML, animations preserved): (1) HFNC slide PEEP unit "3-5 mm H₂O" -> "3-5 cm H₂O" (airway pressures are cm H₂O); (2) Case 3 spelling "non-rebreathrer" -> "non-rebreather"; (3) Oxymizer citation page range "381=383" -> "381-383".

Context added (source-cited): ESICM ARDS 2023 (HFNC first-line for AHRF); ROX index (Roca AJRCCM 2019); awake-proning 2023 rapid practice guideline; RECOVERY-RS (JAMA 2022). FLORALI/Hernandez already cited on deck.

No companion handout is published for this lesson (single interactive board). Medical content should be clinician-reviewed before teaching.